In [1]:
```
#Anabela Petrushevikj
#D213 Task 2
```

In [2]:
```
#Libraries
import os
import re
import json
import random
import string
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, ConfusionMatrixDisplay
)
import tensorflow as tf
from tensorflow.keras.preprocessing.text import Tokenizer
from tensorflow.keras.preprocessing.sequence import pad_sequences
from tensorflow.keras import layers, models
from tensorflow.keras.callbacks import EarlyStopping
```

In [3]:
```
#Reproducibility
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)
```

In [4]:
```
#File path
AMAZON_PATH = "amazon_cells_labelled_D213.txt"
IMDB_PATH   = "imdb_labelled_D213.txt"
YELP_PATH   = "yelp_labelled_D213.txt"
```

In [5]:
```
#Output folder
OUTPUT_DIR = "output"
os.makedirs(OUTPUT_DIR, exist_ok=True)
```

In [6]:
```
#Load and merge datasets
def load_ucisls_file(path: str, source_name: str) -> pd.DataFrame:
    rows = []
    with open(path, "r", encoding="utf-8", errors="replace") as f:
        for line in f:
            line = line.strip()
            if not line:
                continue

            # Most lines are: sentence \t label
            parts = line.split("\t")
            if len(parts) < 2:
                # If a malformed line exists, skip it safely
                continue

            text = "\t".join(parts[:-1]).strip()
            label = parts[-1].strip()
```

```
            # Ensure label is 0/1
            if label not in {"0", "1"}:
                continue

            rows.append((text, int(label), source_name))

    return pd.DataFrame(rows, columns=["text", "label", "source"])


df_amazon = load_ucisls_file(AMAZON_PATH, "amazon")
df_imdb   = load_ucisls_file(IMDB_PATH, "imdb")
df_yelp   = load_ucisls_file(YELP_PATH, "yelp")

df = pd.concat([df_amazon, df_imdb, df_yelp], ignore_index=True)

print("Merged dataset shape:", df.shape)
display(df.head())
```

```
Merged dataset shape: (3000, 3)
```

	text	label	source
0	So there is no way for me to plug it in here i...	0	amazon
1	Good case, Excellent value.	1	amazon
2	Great for the jawbone.	1	amazon
3	Tied to charger for conversations lasting more...	0	amazon
4	The mic is great.	1	amazon

In [7]:

```
#Quick label distribution
print("Label distribution (counts):")
display(df["label"].value_counts().sort_index())

print("\nLabel distribution (percentages):")
display((df["label"].value_counts(normalize=True) * 100).round(2).sort_index())

print("\nSource distribution (counts):")
display(df["source"].value_counts())

print("\nSource distribution (percentages):")
display((df["source"].value_counts(normalize=True) * 100).round(2))
```

```
Label distribution (counts):
label
0    1500
1    1500
Name: count, dtype: int64
Label distribution (percentages):
label
0    50.0
1    50.0
Name: proportion, dtype: float64
Source distribution (counts):
```

```
source
amazon    1000
imdb      1000
yelp      1000
Name: count, dtype: int64
Source distribution (percentages):
source
amazon    33.33
imdb      33.33
yelp      33.33
Name: proportion, dtype: float64
```

In [8]:

```python
#Data Cleaning / Normalization
def detect_unusual_characters(text: str) -> bool:
    #Returns True if text contains characters outside basic ASCII range.
    #Used to identify emojis or non-English symbols for EDA.
    return any(ord(ch) > 127 for ch in text)

#Identify unusual (non-ASCII) rows
df["has_unusual_chars"] = df["text"].apply(detect_unusual_characters)
unusual_count = df["has_unusual_chars"].sum()

print(f"Rows with unusual (non-ASCII) characters: {unusual_count} / {len(df)}")

if unusual_count > 0:
    print("\nExamples with unusual characters:")
    display(df.loc[df["has_unusual_chars"], "text"].head(5))


def clean_text(text: str) -> str:
    text = text.strip()

    #Remove URLs
    text = re.sub(r"http\S+|www\.\S+", "", text)

    #Lowercase
    text = text.lower()

    #Replace non-breaking spaces
    text = text.replace("\u00a0", " ")

    #Remove punctuation
    text = text.translate(str.maketrans("", "", string.punctuation))

    #Remove digits
    text = re.sub(r"\d+", "", text)

    #Remove non-ASCII characters
    text = text.encode("ascii", errors="ignore").decode()

    #Collapse extra whitespace
    text = re.sub(r"\s+", " ", text).strip()

    return text


df["text_clean"] = df["text"].apply(clean_text)
```

```
Rows with unusual (non-ASCII) characters: 17 / 3000

Examples with unusual characters:
1018    It's practically perfect in all of them ▯ a tr...
1080    I'm glad this pretentious piece of s*** didn't...
1178                    The script is
was there a script?
1182            I'll even say it again ▯ this is torture.
1424    This show is made for Americans - it is too st...
Name: text, dtype: object
```

In [9]:

```
#Remove any empty rows
before = len(df)

df = df[df["text_clean"].str.len() > 0].copy()

after = len(df)

print(f"Removed {before - after} empty rows after cleaning.")
print(f"New dataset shape: {df.shape}")
```

```
Removed 2 empty rows after cleaning.
New dataset shape: (2998, 5)
```

In [10]:

```
#Remove duplicates
before = len(df)

df = df.drop_duplicates(subset=["text_clean", "label"]).copy()

after = len(df)

print(f"Removed {before - after} duplicate rows.")
print(f"New dataset shape: {df.shape}")
```

```
Removed 30 duplicate rows.
New dataset shape: (2968, 5)
```

In [11]:

```
#Save cleaned merged dataset
cleaned_path = os.path.join(OUTPUT_DIR, "merged_cleaned_dataset.csv")

df[["text_clean", "label", "source"]].to_csv(cleaned_path, index=False)

print(f"Saved cleaned dataset to: {cleaned_path}")
```

```
Saved cleaned dataset to: output\merged_cleaned_dataset.csv
```

In [12]:

```
#EDA: Vocabulary Size, Sequence Lengths, Embedding Length Proposal
tokens = []
lengths = []

for text in df["text_clean"]:
    words = text.split()
    tokens.extend(words)
    lengths.append(len(words))

#Vocabulary size (EDA estimate)
vocab_size_basic = len(set(tokens))
print(f"Estimated vocabulary size (whitespace tokenization): {vocab_size_basic}")

#Sentence length statistics
lengths = np.array(lengths)
```

```python
stats = {
    "min": int(lengths.min()),
    "25%": float(np.percentile(lengths, 25)),
    "median": float(np.percentile(lengths, 50)),
    "75%": float(np.percentile(lengths, 75)),
    "90%": float(np.percentile(lengths, 90)),
    "95%": float(np.percentile(lengths, 95)),
    "max": int(lengths.max())
}

print("\nSentence length stats (in words):")
for k, v in stats.items():
    print(f"{k}: {v}")

vocab_size_basic, stats
```

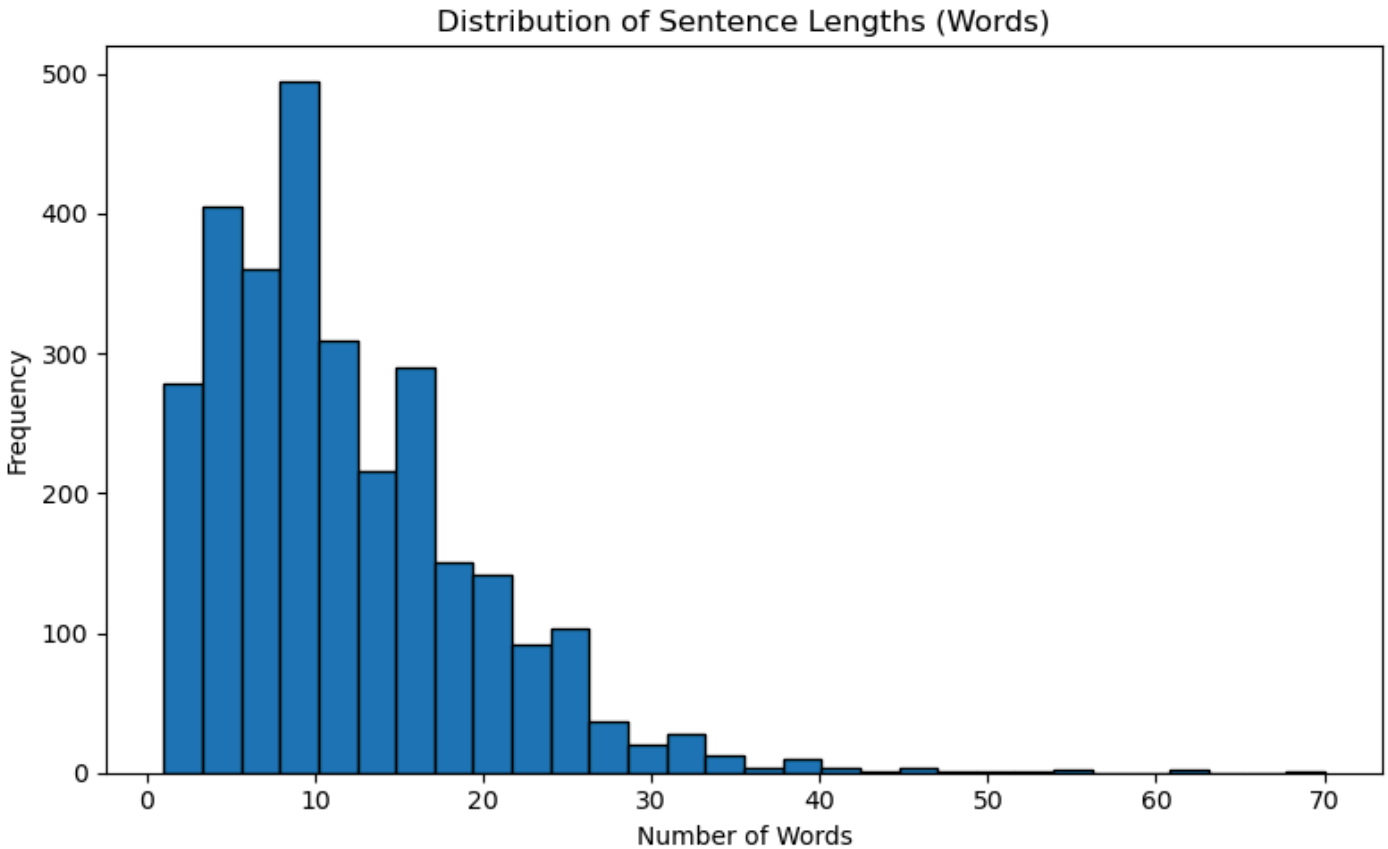
```
Estimated vocabulary size (whitespace tokenization): 5301

Sentence length stats (in words):
min: 1
25%: 6.0
median: 10.0
75%: 16.0
90%: 22.0
95%: 25.0
max: 70
```

Out[12]:

```
(5301,
 {'min': 1,
  '25%': 6.0,
  'median': 10.0,
  '75%': 16.0,
  '90%': 22.0,
  '95%': 25.0,
  'max': 70})
```

In [13]:

```python
#Plot sentence length distribution
plt.figure(figsize=(8, 5))
plt.hist(lengths, bins=30, edgecolor="black")
plt.title("Distribution of Sentence Lengths (Words)")
plt.xlabel("Number of Words")
plt.ylabel("Frequency")
plt.tight_layout()
plt.show()
```



In [14]:
```
#Maximum Sequence Length 95th percentile
max_seq_len = int(np.percentile(lengths, 95))

max_seq_len = max(max_seq_len, 5)

print(f"Chosen MAX_SEQUENCE_LENGTH (95th percentile): {max_seq_len}")
print("This covers approximately 95% of all sentences in the dataset.")
```

```
Chosen MAX_SEQUENCE_LENGTH (95th percentile): 25
This covers approximately 95% of all sentences in the dataset.
```

In [15]:
```
#Embedding length:
EMBEDDING_DIM = 64

print(f"Selected EMBEDDING_DIM: {EMBEDDING_DIM}")
```

```
Selected EMBEDDING_DIM: 64
```

In [16]:
```
#Tokenization
MAX_VOCAB_SIZE = 6000

#Token used for words not seen during training
OOV_TOKEN = "<OOV>"

tokenizer = Tokenizer(
    num_words=MAX_VOCAB_SIZE,
    oov_token=OOV_TOKEN
)

#Fitting tokenizer on cleaned training text
tokenizer.fit_on_texts(df["text_clean"].tolist())
```

In [17]:
```
#Actual vocab used by tokenizer
word_index = tokenizer.word_index

actual_vocab_size = min(MAX_VOCAB_SIZE, len(word_index) + 1)

print(f"Total unique tokens found: {len(word_index)}")
print(f"Actual vocabulary size used in model: {actual_vocab_size}")
```

```
Total unique tokens found: 5302
Actual vocabulary size used in model: 5303
```

In [18]:
```
#Convert to sequences
sequences = tokenizer.texts_to_sequences(df["text_clean"].tolist())

print(f"Number of sequences created: {len(sequences)}")
print("Example sequence (before padding):")
print(sequences[0])
```

```
Number of sequences created: 2968
Example sequence (before padding):
[28, 48, 6, 57, 117, 13, 72, 7, 364, 8, 12, 66, 12, 2, 179, 632, 4, 76, 61, 5, 2185]
```

In [19]:
```
#Padding
PADDING_TYPE = "post"
TRUNC_TYPE   = "post"

X = pad_sequences(
    sequences,
    maxlen=max_seq_len,
    padding=PADDING_TYPE,
    truncating=TRUNC_TYPE
)

#Target variable
y = df["label"].values.astype(np.int32)

print(f"Padded feature matrix shape: {X.shape}")
print(f"Label vector shape: {y.shape}")
```

```
Padded feature matrix shape: (2968, 25)
Label vector shape: (2968,)
```

In [20]:
```
#Single padded sequence:
sample_index = 0

print("Example padded sequence (index 0):")
print(X[sample_index])
```

```
Example padded sequence (index 0):
[  28   48    6   57  117   13   72    7  364    8   12   66   12    2
  179  632    4   76   61    5 2185    0    0    0    0]
```

In [21]:
```
#Train / Validation / Test Split (70/15/15)
#First split: 70% train, 30% temporary
X_train, X_temp, y_train, y_temp = train_test_split(
    X,
    y,
    test_size=0.30,
    random_state=SEED,
    stratify=y
```

```
)

#Second split: 15% validation, 15% test
X_val, X_test, y_val, y_test = train_test_split(
    X_temp,
    y_temp,
    test_size=0.50,
    random_state=SEED,
    stratify=y_temp
)

print("Dataset split sizes:")
print(f"Train: {X_train.shape}")
print(f"Validation: {X_val.shape}")
print(f"Test: {X_test.shape}")
```

```
Dataset split sizes:
Train: (2077, 25)
Validation: (445, 25)
Test: (446, 25)
```

In [22]:

```
#Save prepared dataset
prepared_dataset_path = os.path.join(OUTPUT_DIR, "prepared_dataset.csv")

df[["text_clean", "label", "source"]].to_csv(prepared_dataset_path, index=False)
print(f"\nSaved prepared dataset to: {prepared_dataset_path}")
```

```
Saved prepared dataset to: output\prepared_dataset.csv
```

In [23]:

```
#Identify Sentiment Categories
NUM_CLASSES = 2  # Binary classification: Negative (0), Positive (1)
FINAL_ACTIVATION = "sigmoid"  # Appropriate for binary classification

print(f"Number of sentiment categories: {NUM_CLASSES}")
print(f"Final layer activation function: {FINAL_ACTIVATION}")
```

```
Number of sentiment categories: 2
Final layer activation function: sigmoid
```

In [24]:

```
#Model Hyperparameters
EMBEDDING_DIM = 64
RNN_UNITS = 32
DROPOUT = 0.5

print("Model Hyperparameters:")
print(f"Embedding dimension: {EMBEDDING_DIM}")
print(f"LSTM units: {RNN_UNITS}")
print(f"Dropout rate: {DROPOUT}")


#Build Neural Network
model = models.Sequential([
    layers.Input(shape=(max_seq_len,)),

    #Embedding layer learns dense vector representations of words
    layers.Embedding(
        input_dim=actual_vocab_size,
        output_dim=EMBEDDING_DIM
    ),

    #Bidirectional LSTM captures context from both directions
```

```
    layers.Bidirectional(
        layers.LSTM(RNN_UNITS, return_sequences=False)
    ),

    #Dropout helps reduce overfitting
    layers.Dropout(DROPOUT),

    #Dense layer for nonlinear transformation
    layers.Dense(16, activation="relu"),

    #Output layer: sigmoid for binary classification
    layers.Dense(1, activation="sigmoid")
])


#Compile Model
model.compile(
    loss="binary_crossentropy",
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-3),
    metrics=["accuracy"]
)
```

```
Model Hyperparameters:
Embedding dimension: 64
LSTM units: 32
Dropout rate: 0.5
```

In [25]:

```
#Model summary
print("\nModel Summary:")
model.summary()
```

```
Model Summary:
```

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 25, 64)	339,392
bidirectional (Bidirectional)	(None, 64)	24,832
dropout (Dropout)	(None, 64)	0
dense (Dense)	(None, 16)	1,040
dense_1 (Dense)	(None, 1)	17

Total params: 365,281 (1.39 MB)

Trainable params: 365,281 (1.39 MB)

Non-trainable params: 0 (0.00 B)

In [26]:

```
#Train with Stopping Criteria
early_stop = EarlyStopping(
    monitor="val_loss",
    patience=3,
    restore_best_weights=True
)

EPOCHS = 20
```

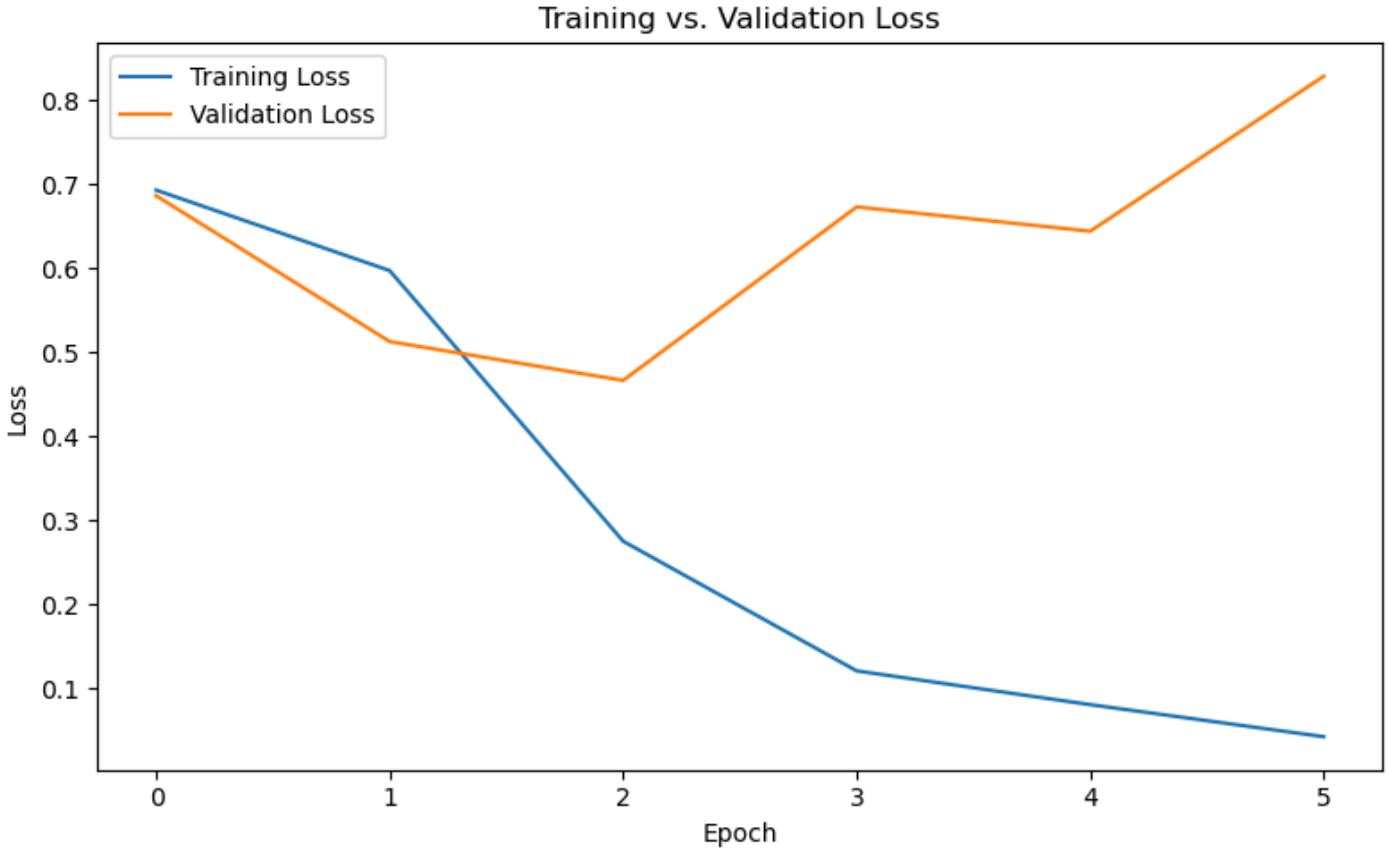
```
BATCH_SIZE = 32

history = model.fit(
    X_train, y_train,
    validation_data=(X_val, y_val),
    epochs=EPOCHS,
    batch_size=BATCH_SIZE,
    callbacks=[early_stop],
    verbose=1
)

#Display final epoch reached
final_epoch = len(history.history["loss"])
print(f"\nTraining stopped after {final_epoch} epochs due to early stopping.")
```

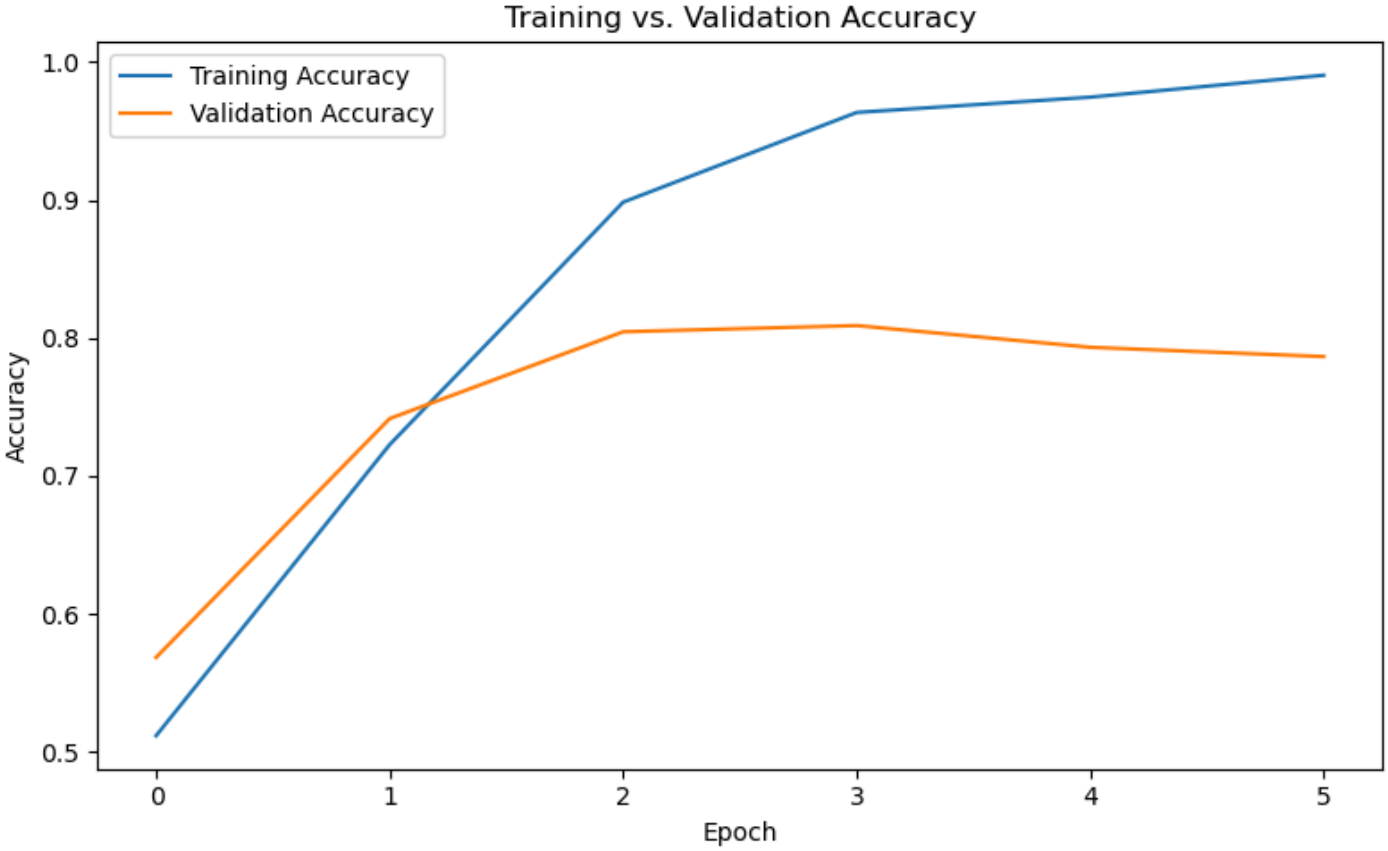
```
Epoch 1/20
65/65 ━━━━━━━━━━━━━━━━━━━━ 18s 69ms/step - accuracy: 0.5118 - loss: 0.6928 - val_accuracy: 0.5685
- val_loss: 0.6861
Epoch 2/20
65/65 ━━━━━━━━━━━━━━━━━━━━ 2s 34ms/step - accuracy: 0.7227 - loss: 0.5970 - val_accuracy: 0.7416
- val_loss: 0.5126
Epoch 3/20
65/65 ━━━━━━━━━━━━━━━━━━━━ 2s 34ms/step - accuracy: 0.8984 - loss: 0.2751 - val_accuracy: 0.8045
- val_loss: 0.4663
Epoch 4/20
65/65 ━━━━━━━━━━━━━━━━━━━━ 2s 34ms/step - accuracy: 0.9634 - loss: 0.1208 - val_accuracy: 0.8090
- val_loss: 0.6728
Epoch 5/20
65/65 ━━━━━━━━━━━━━━━━━━━━ 2s 34ms/step - accuracy: 0.9745 - loss: 0.0807 - val_accuracy: 0.7933
- val_loss: 0.6439
Epoch 6/20
65/65 ━━━━━━━━━━━━━━━━━━━━ 3s 40ms/step - accuracy: 0.9904 - loss: 0.0425 - val_accuracy: 0.7865
- val_loss: 0.8282

Training stopped after 6 epochs due to early stopping.
```

In [27]:

```
#Visualizations: Loss and Accuracy
loss = history.history["loss"]
val_loss = history.history["val_loss"]
acc = history.history["accuracy"]
val_acc = history.history["val_accuracy"]

#Plot: Loss
plt.figure(figsize=(8, 5))
plt.plot(loss, label="Training Loss")
plt.plot(val_loss, label="Validation Loss")
plt.title("Training vs. Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.tight_layout()
plt.show()

#Plot: Accuracy
plt.figure(figsize=(8, 5))
plt.plot(acc, label="Training Accuracy")
plt.plot(val_acc, label="Validation Accuracy")
plt.title("Training vs. Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
```

```
plt.legend()
plt.tight_layout()
plt.show()
```





In [28]:
```
#Evaluate Model on Test Set
#Evaluate overall performance
test_loss, test_acc = model.evaluate(X_test, y_test, verbose=0)
```

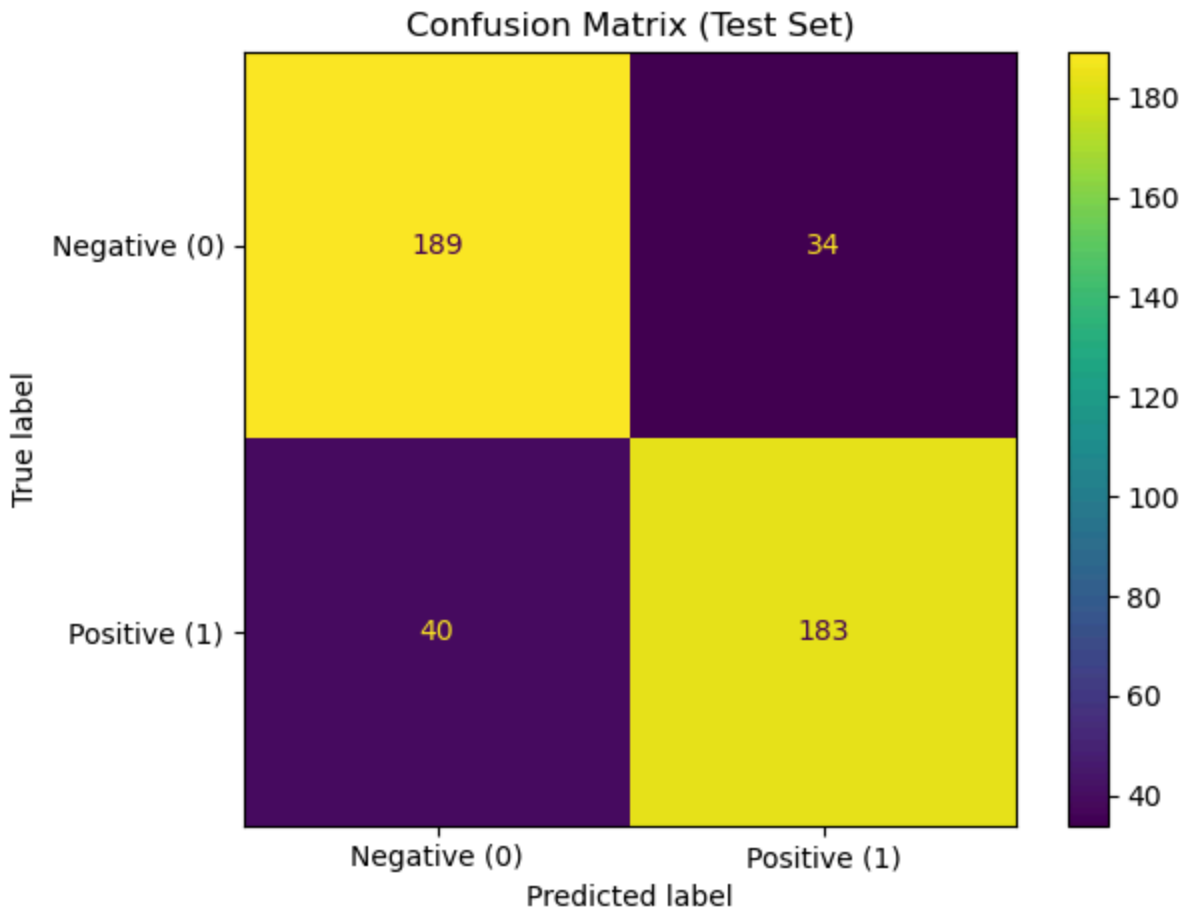
```python
print("Test Set Performance:")
print(f"Loss: {test_loss:.4f}")
print(f"Accuracy: {test_acc:.4f}")

#Generate predictions
y_prob = model.predict(X_test).ravel()
y_pred = (y_prob >= 0.5).astype(int)

#Classification metrics
accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, zero_division=0)
recall = recall_score(y_test, y_pred, zero_division=0)
f1 = f1_score(y_test, y_pred, zero_division=0)

print("\nDetailed Classification Metrics:")
print(f"Accuracy : {accuracy:.4f}")
print(f"Precision: {precision:.4f}")
print(f"Recall   : {recall:.4f}")
print(f"F1 Score : {f1:.4f}")
```

```
Test Set Performance:
Loss: 0.4382
Accuracy: 0.8341
14/14 ━━━━━━━━━━━━━━━━━━━━ 3s 113ms/step

Detailed Classification Metrics:
Accuracy : 0.8341
Precision: 0.8433
Recall   : 0.8206
F1 Score : 0.8318
```

In [29]:

```python
#Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["Negative (0)", "Positive (1)"]
)

plt.figure(figsize=(6, 6))
disp.plot(cmap="viridis", values_format="d")
plt.title("Confusion Matrix (Test Set)")
plt.tight_layout()
plt.show()
```

```
<Figure size 600x600 with 0 Axes>
```



In [30]:
```
#Saving the Trained Model
model_save_path = os.path.join(OUTPUT_DIR, "sentiment_bilstm_model.keras")

model.save(model_save_path)

print(f"Model location: {model_save_path}")
```

```
Model location: output\sentiment_bilstm_model.keras
```

In [ ]:

In [ ]: